

MARCO TAN

Technology Consultant

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SUMMARY

Mechatronics and Biomedical Engineering student at McMaster University. Experience in embedded, web, and machine learning development in professional environments and project settings. Seeking software internship and co-op positions for the summer of 2025.

SKILLS

Languages: C, C++, Rust, MatLab, Maple, Lua, Python, TypeScript

Technologies: GCP, Firebase, Next.js, TailwindCSS, PyTorch, Express, Flask, Fastify, SolidWorks, Autodesk Inventor, KiCAD

EXPERIENCE

- 10/2023 – Present **Technology Consultant** **LocalStudent**
- Spearheaded development of an AI dispute model dataset containing thousands of image and tabular data entries. Images and data were stored using Firebase Storage and Firestore.
 - Led a team of interns on projects to optimize and refine key operational processes and workflows with services like our CRM and data collection infrastructure, used for customer/contractor interactions.
 - Maintained critical operational infrastructure such as Twilio, Zapier automations, 3CX, GCP cloud infrastructure, and third-party platforms and services.
- 6/2021 – 7/2022 **Mentor** **Brampton Robotics**
- Taught multiple teams of 4+ students C/C++ with the V5 system and PROS real-time operating system for control of VEX robots.
 - Introduced introductory control theory, PID, feedforward systems, odometry, and sensor filtering to teams to improve the control code they wrote.
 - Introduced modelling in Autodesk Inventor to hardware design teams to improve their design and iteration times.
 - Prepared all competition teams by helping to refactor codebases and testing hardware systems.

EXTRACURRICULARS

- 2024 - Present **McMaster iGEM - Dry Lab Member** **McMaster University**
- Led research into different machine learning models for protein-ligand affinity binding, including CNNs, Bayesian networks, and transformer-based models.
 - Conducted analysis of data sets used in model training with Mahalanobis distances and statistical methods.
 - Helped design an evolutionary algorithm to engineer proteins with higher binding affinities, with ML-based fitness functions and folding simulations.

EDUCATION

- 2022 - 2027 **BEngBME - Mechatronics and Biomedical Engineering** **McMaster University**
- Enrolled full-time. CGPA 3.7. Dry laboratory member of McMaster iGEM. Took courses in software development, algorithms and abstract data types, analog and digital circuits, and design.

PROJECTS

- Web Development **Personal Website v4.1** <https://github.com/IonicArgon/web-v4.1>
- Built a new personal website because the old one didn't satisfy me. Built with Next.js, TailwindCSS, and Firebase. Hosted on Vercel.
- Robotics **VEX 2022 Competition Robot Codebase** <https://github.com/Discobots-1104A/2021-2022-branch-sample>
- Codebase containing teleop and autonomous code routines for control of robot during VEX VRC competition matches. PID and feedforward control for autonomous movement and controlled actuator manipulation. Written in C++ and PROS RTOS.
- Algorithms **Linear Systems Solver in C** <https://github.com/IonicArgon/LinearSystems>
- C implementation of a CSR Matrix Market file reader and linear system solver with Jacobi iteration and Successive over-relaxation methods. Permutation preconditioning of systems to better optimize convergence.
- Full Stack **HeartFlow - Pacemaker Firmware and Control Software** <https://github.com/andrewderango/HeartFlow>
- Simulink-based firmware for FRDM K64F-based pacemaker. Device controller-monitor software developed with Electron + React, serial UART communication w/ Python backend. Developed with model-driven engineering and safety engineering design principles.